In [7]:
```
import numpy as np
```

In [8]:
```
def get_loss(y_true: np.float32, y_pred: np.float32) -> np.float32:
    return np.square(y_true - y_pred)

def backpropagation(y_true: np.float32, y_pred: np.float32) -> np.float32:
    return -2 * (y_true - y_pred)
```

Task 1

In [9]:
```
y_true, y_pred = 1, 0.6

L = get_loss(y_true, y_pred)
dy_dx = backpropagation(y_true, y_pred)

print(f"dL/dŷ = {dy_dx}")
print(f"")
```

```
dL/dŷ = -0.8
```

Task 2

In [10]:
```
w, x = 0.5, 2

dl_dw = dy_dx * x

print(f"dy/dw = {x}")
print(f"dL/dw = {dl_dw}")
```

```
dy/dw = 2
dL/dw = -1.6
```